DocScout: Query-Aware Fine-Grained Evidence Retrieval from Long Documents

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Problem Statement

Question answering over long documents usually returns a whole PDF or page, or a generated answer that cannot be checked. Users need the exact paragraph, table cell or figure that answers the question, with its location and a reason it was retrieved.

Objective

Retrieve fine-grained, query-aware evidence (paragraph, table cells, figure) within a long document, with provenance (page, box, scores, graph path, why-line), and produce a short answer grounded only in that evidence with validated citations; let users refine the ranking with relevance feedback.

Dataset Used

MP-DocVQA (lmms-lab/MP-DocVQA, val): 49 documents, 250 pages, 293 questions, 2,976 parsed components; split by document into DEV (10 docs, 76 q) and TEST (39 docs, 217 q). Qrels: answer page and the components containing a gold answer (94.9% coverage). Demo: US Census P60-281 report (31 pages, 344 components).

Methods Used / Algo Used

- Docling parsing (layout, TableFormer, RapidOCR) into typed components; table/figure text serialization; document structure graph (networkx).
- Three branch indexes: BM25 + BGE-base-en-v1.5 dense (FAISS) for text, tables, figures; CLIP ViT-B/32 for figure images.
- Query-type classifier (rules + gpt-oss-20b when unsure) setting soft branch weights; min-max normalization; weighted-sum fusion (DEV-tuned).
- Cross-encoder rerank (ms-marco-MiniLM-L-6-v2, top 30); type-conditional modality slots; table cell locator; graph context expansion.
- Grounded LLM answering (gpt-oss-20b, JSON, citation validation, "Not found"); provenance why-line.
- Relevance feedback: session Rocchio (q' = alpha q + beta mean(relevant) - gamma mean(non-relevant), re-fuse, rerank) and a capped cross-session memory prior; FastAPI + React UI.

Results (Metrics Used)

Final TEST numbers (eval/results/summary.md, P6/P7 reports):

- **Component R@1:** 0.488 (dense BGE baseline 0.362)
- **Component MRR:** 0.653 (BGE 0.544; paired t-test $p < 0.0001$)
- **Component nDCG@5:** 0.644
- **Page R@1 / MRR (documents with >= 2 pages):** 0.733 / 0.828 (random 0.270 / 0.501); page nDCG@5: not recorded (nDCG@10 0.869)
- **ANLS, baseline (top-1 evidence):** 0.486 [0.424, 0.546]
- **ANLS, final (top-3 + context + cells):** 0.646 [0.585, 0.707]; +0.161 [+0.103, +0.218] over baseline
- **ANLS, oracle (gold page, 40 q):** 0.791
- **Rocchio residual MRR:** 0.312 vs 0.289 without feedback (+0.023 [-0.048, +0.091]): **not significant on TEST**
- **Query classifier accuracy:** 0.751; **table-cell locator hit rate:** 0.548